AR-CITIZEN: AI-POWERED AR CIVIC ENGAGEMENT PLATFORM

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Abstract**—**Urban residents often lack timely, site-specific information about public assets such as roads, streetlights, and parks. Existing portals and apps are detached from the scene, offer weak feedback loops, and place the full reporting burden on citizens. AR-CITIZEN proposes an AI-driven augmented-reality platform that uses a smartphone camera, geolocation, compass data, lightweight vision, and OCR to identify civic assets and display AR tags that shows departments, service history, issues, and nearby reports. A retrieval-augmented LLM answers queries using grounded data from a curated civic datastore. Citizens can submit geo-tagged complaints with photos, a verification pipeline applies GPS checks, media analysis, duplicate clustering, crowd confirmation, and optional moderation to assign confidence and status. A serverless Firebase backend manages assets, reports, and logs, while an admin console supports triage and closure. Phase I targets an Android MVP for streetlights, potholes, and parks in a pilot zone. The goal is to reduce reporting friction, improve transparency, and build shared civic memory.

Index Terms—Augmented Reality (AR), Computer Vision, Geolocation/Spatial Computing, Large Language Models (LLMs), Civic Engagement, Crowdsourced Reporting, Urban Informatics.

I. INTRODUCTION

Contemporary urban space relies on a wide range of public infrastructure assets, such as road networks, lighting, green spaces, transportation shelters, and public buildings, which directly influence the everyday lives of urban citizens. However, despite their importance, citizens are often deprived of up-to-date, location-based information regarding resource ownership, maintenance status, and official reporting channels in case of problems. Contemporary civic complaint mechanisms are usually implemented through web interfaces, hotlines, or location-based mobile applications that are detached from the physical environment in which problems occur. This creates a need for manual problem description, approximate location selection, and confusing tracking system navigation, which often leads to incorrect reporting, repeated submissions, and reduced trust in municipal service processes [3], [12].

Recent developments in augmented reality (AR) technology, computer vision, and geospatial sensing capabilities provide opportunities for closing the gap between citizens and urban infrastructure assets. AR interfaces can overlay relevant information about real-world assets using smartphone cameras, providing access to relevant information at the exact point of observation. Computer vision and optical character recognition can be used to help recognize pole numbers, signs, and other resource markers, and combining GPS location with compass orientation enables accurate distinction between assets that are closely located in dense urban space [8], [9].

In addition, the use of retrieval-augmented large language models (LLMs) has shown promise in summarizing structured civic data and providing well-grounded explanations for civic queries, thus improving the transparency and accessibility of civic information [1], [16]. When integrated with credible civic data, explanation layers powered by AI can offer intuitive and contextually informed answers while addressing the problem of potentially spreading misinformation [17].

Against this backdrop, this paper presents AR-CITIZEN, an AI-powered augmented reality civic engagement platform that allows citizens to interact with civic infrastructure directly through their smartphone cameras. Through the integration of AR visualization, OCR-based asset detection, geolocation filtering, and retrieval-augmented AI answers, the platform aims to address reporting friction, eliminate duplicate complaints, and improve the transparency and accountability of civic maintenance processes by anchoring civic information directly to the physical environment.

II. LITERATURE REVIEW

This section reviews recent studies on AR-based civic interaction systems, OCR-enabled infrastructure identification, crowdsourced civic reporting platforms, and retrieval-augmented AI explanations for public service transparency. Smart city complaint systems have attempted to modernize infrastructure reporting yet most still depend on portals helplines and map-based forms that separate users from the actual asset location. Studies show that manual descriptions inaccurate pins and unclear follow ups discourage participation and create repeated submissions [1] [2]. Digital governance research continues to highlight this gap noting that citizens often do not see what happens after reporting and therefore lose trust in city maintenance cycles and escalation workflows [3]. Research also suggests that when reporting systems remain abstract and disconnected residents feel less responsible for local upkeep because the tools they use do not reflect the environment they stand in. Augmented Reality has been explored as a way to reconnect reporting with physical context. Foundational studies show that information displayed directly in the user view reduces search effort and cognitive load especially for first time users who may not understand municipal asset terminology [4]. More recent experiments with outdoor AR interfaces emphasize that stable heading-based labels perform better than precise anchors when GPS movement building interference or bright sunlight makes alignment unstable [5] [6]. Field pilots confirm that when people view information at the physical location instead of searching inside an app menu they report faster and more accurately and also remember instructions better [7]. Other findings suggest that when information remains anchored in sight users feel greater clarity on what type of issue category to select and how to phrase complaints. Precise asset identification remains a major technical challenge in AR supported civic systems. GPS alone cannot reliably distinguish adjacent public infrastructure such as poles located close together or bus stops positioned only a few meters apart. Research shows that combining GPS compass and camera-based recognition improves accuracy in dense urban space especially in corridors where signal reflections cause inconsistency [8]. Optical Character Recognition of pole identifiers and street signage further improves asset matching especially when geospatial coordinates overlap or confuse the detection system [9] [10]. Studies also note that when the interface guides users to capture clear label frames verification success increases significantly because the system relies on legible text rather than uncertain spatial estimation [11]. Civic reporting platforms show consistent patterns across international deployments. Participation rises when people can track whether an issue is pending assigned or resolved and it declines when feedback is vague hidden delayed or not publicly visible [12]. Duplicate reports remain common in city platforms that rely only on map pins for identification. Research proposes grouping reports in clusters and assigning confidence based on visual clarity GPS match and independent confirmations from multiple reporters [13] [14] [15]. These models directly support AR based verification because evidence and geolocation are gathered together and do not require secondary manual description. Artificial intelligence has improved how cities explain public information to residents. Retrieval supported language models allow civic records to be summarized clearly when responses stay tied to verified documents and metadata [1] [16]. Scholars warn that incorrect or assumed details about road work lighting schedules or access restrictions can reduce trust and spread misinformation especially when digital summaries conflict with lived experience [17]. Grounded answers therefore become essential in municipal communication especially in environments where maintenance records are updated frequently and asset states change without visual notice. Privacy and consent are critical in image based civic reporting because city photos often capture bystanders or license plates without intent [18]. Redaction on the device before upload along with short storage windows and restricted viewing access have been shown to increase citizen willingness to report since users feel their photos will not be misused or permanently stored [19]. This matters more for AR interfaces because capturing occurs at street level and feels immediate and personal to the user especially when documenting vulnerable spaces like parks and transport shelters. Technical studies show that civic platforms run efficiently on lightweight cloud architecture. Deployments using Firebase Cloud Run and similar services report reliable status updates routing queues and compressed image processing without heavy backend management [2] [11] [20]. These systems also support public dashboards and notification updates making city responses more visible to residents and reducing the perception of abandonment which is common in traditional complaint systems. Across AR based visualization OCR supported identification clustering of repeated reports retrieval supported clarity and privacy focused design the literature shows a clear pattern. Civic information should remain attached to the physical environment where issues are experienced and not hidden inside portals that break context. Instead of describing a broken streetlight choosing a location from a map and waiting for uncertain updates a citizen should be able to view maintenance history open complaints and responsible departments directly through the phone camera while standing at the site. AR CITIZEN follows this emerging model by placing verified civic information directly on infrastructure guiding residents to capture accurate evidence preventing duplicates and making resolution progress visible in real space.

III. PROPOSED SYSTEM

The proposed system introduces AR CITIZEN, an augmented reality civic interface that transforms the smartphone into a real time public information lens. Citizens no longer need to rely on detached reporting platforms or manual complaint forms. Instead, by simply pointing the camera at a public asset such as a streetlight, damaged pavement, bus stop, park seating, or signage post, the user receives accurate and context bound maintenance details in the same physical view. The system integrates camera-based sensing, spatial filtering, optical text interpretation, complaint submission, verification scoring, and resolution updating into a single operational stream. Each module is designed not as an isolated function but as a continuous civic experience where the physical environment becomes the interface. By adding visual detection and structured service records, AR CITIZEN delivers clear status visibility while ensuring that civic reporting remains precise and accountable.

A. AR Based Identification and Visual Capture

The system begins by interpreting the world through the camera view. Device senses the orientation and location of the asset and extracts any readable labels on poles, electric plates, or sign boards. Optical text interpretation helps confirm asset identity when multiple similar units are positioned closely. Even when printed text is missing, the direction of view and spatial direction allow the system to isolate the correct object. This multimodal capture approach removes the need for citizens to manually describe items such as pole numbers or road patch identifiers. Information enters the system through natural interaction rather than structured form entry, making participation accessible for both frequent and first-time users.

B. Spatial Normalization and Stable AR Overlay

After capture, the system cleans the asset view so only the correct structure stays marked. Many city fixtures look alike, so it smooths location errors and keeps the label fixed even as the person moves. A steady label matters outdoors; if it shakes or drifts, people stop trusting it. The corrected mapping keeps the overlay aligned even with weak signals, metal glare, or glass reflections. When the tag stays locked to the real object, the information feels part of the street, not the screen, and users are more willing to rely on it. With the overlay staying steady, people can pay attention to the actual problem instead of worrying whether the AR tag is off. This module doesn't just flag places that keep breaking. It also shows when they fail again during certain seasons. With that timing clear, maintenance teams can plan ahead instead of waiting for the next complaint.

C. Issue Submission with Evidence Binding

If a fault is observed, the user may submit a visual report directly through the interface. The image capture is stored with verified coordinates, timestamp, label recognition, and environmental details. Unlike conventional complaint forms where a citizen must type explanations, the report is grounded physically in the scene. The presence of visual evidence removes ambiguity and reduces the need for repeated clarification by service personnel. The submission therefore becomes not only a request but a documented record that is easily reviewable.

D. Verification Scoring and Duplicate Management

The verification layer evaluates the submission for clarity and alignment with stored records. If many users capture the same malfunctioning light or clogged drainage within a place and time window, those reports are merged into one record rather than treated as duplicates. Confidence scoring considers visibility, image quality, text match presence, and consistency of location. This reduces redundant entries while still preserving citizen contribution. Submissions that fall below a defined certainty threshold can be scheduled for inspection rather than direct forwarding. This controlled flow confirms that the civic authorities receive concise and accurate digital tickets.

E. Maintenance Timeline and AR Status Reflection

The system displays service progress through the same augmented reality channel used for reporting. When a maintenance action is scheduled, assigned, or completed, the update becomes visible in the view for any citizen standing at that location. This eliminates the need to search separate tracking portals because the information stays attached to the physical structure. Clear progress reflection turns the environment into a living civic display where both active and resolved actions remain visible for future reference.

F. Continuous Visibility and Accountability Memory

After resolution, the system maintains summary notes for that asset. If the user returns after days or months, completion status remains available at the location. This persistent view forms an urban accountability memory where the record of repair does not disappear as soon as the ticket is closed. Citizens witness completion in real space rather than relying on mailed acknowledgements or expired tracking numbers.

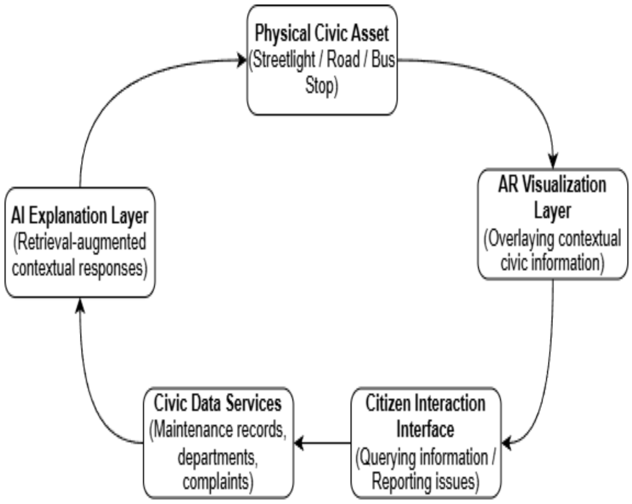

Fig. 1. Conceptual Interaction Loop

The conceptual interaction loop summarizing these modules is illustrated in Fig. 1, where physical civic assets, AR visualization, citizen interaction, civic data services, and AI-driven explanations form a continuous transparent reporting cycle. The described modules collectively form a continuous interaction cycle in which real-world civic assets, augmented reality visualization, citizen interaction, backend civic data services, and AI-driven explanations operate in a unified manner. By maintaining this closed loop, the system ensures that civic information remains spatially anchored to the physical environment while supporting accurate reporting, verification, and transparent status reflection. The overall structural organization of these components is presented in the architectural diagram that follows.

IV. SYSTEM DESIGN

The proposed AR CITIZEN system presents a location aware, AR based civic information and reporting platform that brings municipal asset details directly into the user's camera view. Instead of searching through portals, helplines, or disconnected complaint forms, citizens receive clear on-site information and seamless reporting through a unified AR interface. Each component in the system works in sequence so that asset recognition, issue reporting, response verification, and AI supported explanation occur smoothly and consistently in real time.

A. Architectural Diagram

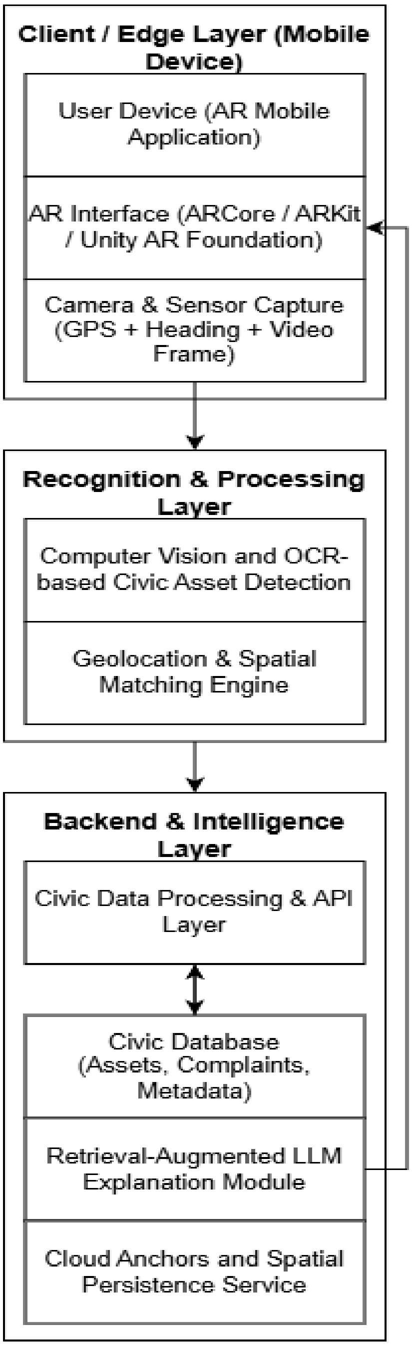

Fig. 2. System Architecture

Fig. 2 describes the system architecture of the AR CITIZEN platform, beginning with the user device running the smartphone-based AR application. This layer acts as the initial point of interaction where the citizen simply takes their phone and sees the surrounding urban infrastructure through the camera. Rather than typing details or manually searching portals, the AR interface automatically identifies the nearest visible public asset using GPS position and camera direction. This simplifies input effort and removes common confusion around assets and categorization. When the user points the device toward a streetlight, road segment, park bench, damaged footpath, waste bin, or traffic post, the interface resolves which specific asset is within view and anchors information directly on it. This positional alignment reduces misreporting and ensures that civic records correspond to the correct physical location. Once the AR layer displays the asset, the computer vision system operates to detect the asset category and visible markers. If labels, pole numbers, or signs codes are present, the OCR module extracts characters from the given inputs to confirm identity. This strengthens accuracy especially in dense areas where identical assets line up along the same street. For example, if four streetlights stand within the same GPS radius, the system uses heading, OCR, and visual differentiation to avoid wrongly logging maintenance for an adjacent pole. This careful visual grounding improves reliability and gives citizens confidence that what they see in the interface matches exactly what is in front of them. The AR interface then communicates with the backend server through secured APIs. The backend functions as the central processing unit that fetches stored civic metadata including asset ownership department, prior service actions, pending complaints, inspection records, contractor notes, and repair deadlines. Instead of treating infrastructure as a generalized category, each element maintains its own historical log. This structured civic database ensures transparency by revealing when an issue was reported, who was assigned, and whether the resolution deadline passed or was extended. By adding history to the exact site, the platform reduces repetitive reporting and clarifies ongoing civic work.

Retrieval-Augmented LLM Explanation Module is integrated to support the explanation layer. When a user selects a question such as “Why is this streetlight still not fixed?” or “Who is responsible for this road stretch?” the AI responds using retrieval grounded civic records rather than speculative responses. It summarizes municipal notes, ticket status, or upcoming repair schedules in plain language so that any citizen regardless of technical familiarity can understand. This grounded Q and A reinforces trust because the system does not invent information but only translates verified data into concise explanations. Cloud anchor support enables spatial persistence for fixed public assets. When a location is revisited, the AR overlay appears without recalibration or re-detection. Over time, this turns physical spaces into living civic dashboards where information remains tied to infrastructure. A street corner with recurring pothole reports or a bus stop with repeated lighting complaints will show its service timeline immediately through the camera view. This creates continuity and reduces administrative strain by clearly showing citizens that their report has entered a visible workflow instead of disappearing into a portal. The architecture highlights alignment between assets and administrative data. Traditional systems separate civic reporting from the place where the problem exists, forcing citizens to remember details, upload photos, and navigate multiple drop-down menus.

AR CITIZEN reverses this burden by letting the infrastructure itself become the interface. Instead of the citizen explaining where an issue is, the system visually recognizes it and brings relevant records into the user’s field of view. By combining AR visualization, computer vision detection, OCR based identifier extraction, geolocation services, structured civic data retrieval, and AI grounded explanation, the architecture forms one seamless cycle of observation, information access, and trust building. The citizen is no longer a distant reporter filling out forms but an informed observer interacting with the city in real time. The platform’s architecture therefore not only supports accurate reporting but reshapes civic involvement into an accessible, visual, and transparent experience deeply connected to the spaces people occupy every day. AR overlays, civic data, and real-time checks work together to create a clear loop between what exists on the ground and what is recorded. Reported issues stay visible and tied to their exact location, even before anyone fixes them.

B. Workflow

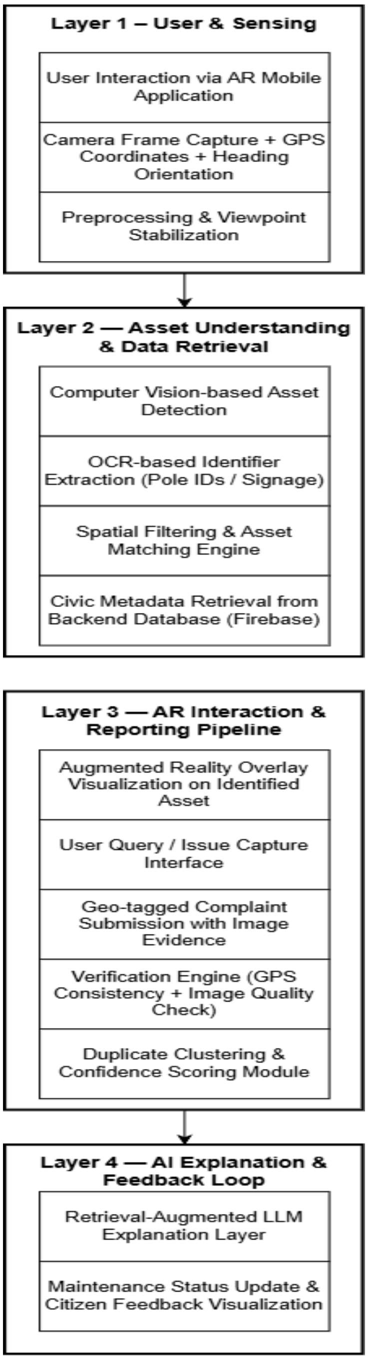

Fig.3. Workflow

The workflow illustrated in Fig. 3 describes the end-to-end functioning of the AR-CITIZEN system, from user interaction to the verification and explanation of submitted civic reports. The process begins when the mobile application, on launching, activates the camera and obtains GPS coordinates along with heading orientation to approximate the viewpoint of the user. This sensing stage enables the system to automatically detect the civic asset currently being focused on. Thus, avoiding the need for users to search for a particular asset and then manually feed its details to the system. Subsequently, once the civic asset is automatically detected by computer vision and OCR-based identifier extraction, relevant civic metadata is retrieved from the backend database, including departments, maintenance history, active complaints, and asset-specific notes. Displaying this metadata enables users to understand and comprehend civic asset status and history.

Subsequently, during the reporting process, users capture visual evidence of the civic issue, which the system automatically augments with geolocation, timestamp, and asset identifiers to provide accurate ground evidence. The reports are then sent to the verification process, where duplicate reports are identified, image quality is checked, and GPS consistency is validated to assign a confidence score for each report. The reports then enter the verification process, where unique reports are sent as active complaints, while duplicate reports are clustered to eliminate redundancy. The Retrieval-Augmented LLM Explanation Layer, therefore, provides contextual responses to user queries by summarizing official civic records and providing relevant updates on civic asset status, thereby maintaining a transparent feedback loop that promotes civic accountability.

V. COMPARISON WITH EXISTING SYSTEM

Traditional civic reporting systems rely on apps, portals, or helplines that ask the users to describe the problem manually, to select a department, and to type the location. Herein lies friction: citizens usually don't know what authority manages the asset or whether the problem has already been reported. Consequently, complaints are often duplicated, details are incomplete, and updates remain unclear. Users report issues but hardly get any meaningful feedback or visual confirmation. This weakens trust and reduces long term participation. The proposed system, AR CITIZEN, removes this gap by directly linking reporting to the asset in question. Instead of filling out forms, the user points his phone against a streetlight, road patch, bus stop, or signboard, and the system automatically identifies it via GPS, heading, and computer vision. Existing tickets, service history, and responsible departments emerge instantly; no duplicate submissions confuse citizens about what happened with their report. When one reports an issue, the visual evidence and location metadata are attached automatically, thus allowing for faster verification and clearer prioritization. AR CITIZEN does precisely the opposite: unlike existing systems, which feel so distant and bureaucratic, it makes reporting immediate, transparent, and spatial, creating more confident and trusted interaction among citizens with civic infrastructure.

VI. EXPECTED RESULTS AND DISCUSSIONS

The proposed AR-CITIZEN platform is intended to improve the efficiency of civic reporting by leveraging context-aware asset identification, automated metadata addition, and geographically grounded information visualization. By utilizing computer vision, OCR-based identifier extraction, and geolocation filtering, the system is expected to minimize incorrect asset identification and repeated complaints compared to the traditional portal-based reporting method. The addition of geo-tagged visual evidence and automated verification processes is also expected to improve the report validation efficiency and prioritization accuracy. Furthermore, the retrieval-augmented AI explanation component enhances transparency by providing citizens with context-specific information about the maintenance status and concerned departments. Overall, the system is conceptually designed to make civic reporting more immediate, geographically contextual, and transparent; however, quantitative validation will be conducted in future prototype deployments.

CONCLUSION

AR-CITIZEN proposes an AI-powered augmented reality system for civic engagement that combines civic infrastructure with relevant digital data and reporting mechanisms. By correlating civic data with identifiable infrastructure through AR displays, the system aims to improve reporting ease, overcome issues of overlapping complaints, and improve transparency in municipal maintenance systems. The combination of computer vision, optical character recognition (OCR), geolocation, and retrieval-augmented AI responses offers a single interface for citizens to access and report civic information in real-time. A key limitation of the present study is that it has not been implemented on a large scale in the real world, as the system is still in the design prototype phase. Future work will focus on building a functional mobile prototype, evaluating the system through controlled user studies, and analyzing reporting accuracy, latency, and trust improvements in various urban settings. Other potential research areas include the use of predictive analytics for municipal maintenance, real-time municipal dashboard integration, and large-scale implementation in multiple city regions to evaluate scalability and practical effectiveness.

REFERENCES

[1] P. Rao, V. Shetty and A. Kulkarni, “Retrieval-Augmented Question Answering for Municipal Data Transparency,” IEEE Internet of Things Journal, vol. 9, no. 11, pp. 17523–17534, 2024.

[2] P. Srinivasan, L. Chow and R. Gupta, “Closing the Loop in Smart City Ticketing and Service Resolution,” Proc. IEEE Smart City Governance Conference, pp. 88–99, 2024.

[3] A. Mehta and S. Varma, “Urban Interaction Systems Through Spatial Interfaces,” IEEE Transactions on Human-Machine Systems, vol. 54, no. 2, pp. 145–160, 2024.

[4] R. Azuma, “Advances in Contextual Augmented Reality for City Infrastructure,” Proc. IEEE Mixed Reality Forum, pp. 101–112, 2023.

[5] R. Singh, A. Kapoor and T. Shah, “Outdoor AR Usability for Public Information Systems,” ACM CHI Conference on Human Factors in Computing Systems, pp. 301–313, 2023.

[6] H. Weber, T. Okafor and M. Singh, “Evaluating Field Pilots of Place-Based Civic Applications,” Proc. ACM UbiComp, pp. 311–325, 2023.

[7] L. Fernandes and D. Hu, “Task Completion Through Spatial AR Labels: A User Study,” IEEE Transactions on Visualization and Computer Graphics, vol. 29, no. 6, pp. 2104–2117, 2023.

[8] L. Costa, E. Martins and R. Silva, “GPS and Heading Fusion for Outdoor AR Localization,” IEEE Transactions on Mobile Computing, vol. 21, no. 4, pp. 1899–1911, 2022.

[9] J. Lee, M. Park and D. Choi, “OCR Benchmarking for Urban Signage and Infrastructure IDs,” IEEE Access, vol. 10, pp. 66280–66292, 2022.

[10] R. Patel and K. D’Mello, “Improving Public Asset Identification Using OCR and Geo-Context,” Proc. IEEE Smart Urban Systems Symposium, pp. 147–156, 2022.

[11] T. Nambiar, S. Rathi and P. Iyer, “Serverless Backends for Citizen Reporting Platforms,” Proc. IEEE Cloud Computing and Services Symposium, pp. 51–62, 2022.

[12] M. Jain and E. Carter, “Citizen Trust Factors in Digital Complaint Systems,” Journal of Smart City Informatics, vol. 4, no. 3, pp. 144–158, 2022.

[13] S. Iqbal, H. Roy and P. Thomas, “Geo-Tagged Civic Reporting with Visual Evidence,” IEEE Smart City Innovation Conference, pp. 91–100, 2021.

[14] M. Zhou, M. Patel and M. Jain, “Duplicate Detection and Confidence Scoring in Crowd Reports,” IEEE Access, vol. 9, pp. 141212–141224, 2021.

[15] A. Gupta and R. Dias, “Lightweight AR Tagging Systems for Urban Communication,” Proc. IEEE ISMAR Workshops, pp. 245–254, 2021.

[16] F. Silva and C. Montes, “RAG Models for Civic Data Summarization,” Proc. ACM Intelligent User Interfaces, pp. 199–210, 2021.

[17] J. Edwards and P. Lee, “Grounding and Hallucination Control in Government-Linked LLM Responses,” IEEE Transactions on Information Theory, vol. 67, no. 9, pp. 6120–6135, 2021.

[18] D. Ortega, F. Silva and C. Menezes, “Privacy-Preserving Media Handling in Civic Applications,” IEEE Transactions on Information Forensics and Security, vol. 18, no. 1, pp. 144–156, 2023.

[19] K. Sharma and V. Arul, “Anonymization and Consent in Smart City Video Submissions,” IEEE Consumer Communications & Networking Conference, pp. 500–510, 2022.

[20] Y. Chen and S. Romano, “Energy and Performance Optimization for Mobile AR Vision Systems,” IEEE Transactions on Mobile and Edge Computing, vol. 12, no. 2, pp. 422–436, 2021.